



APPLICATION NOTE

Setting the I2C SCL duty cycle and an exact bus frequency with the Binho Supernova and Pulsar

AN0015

Document information

Keywords	AN0015, Supernova, Pulsar, I2C, SCL, duty cycle, tHIGH, tLOW, timing margin, bus frequency, pyCosmicSDK
Abstract	Some I2C targets, and some qualification plans, care how each SCL period is split between high and low, not only how many periods there are per second. Adapter firmware 4.6.0 and pyCosmicSDK 1.6.0 let a Binho Supernova or Pulsar set that split directly, from 20 to 80 percent high in steps of 10, and setting it also makes the bus frequency exact rather than rounded. This note explains the call, supplies a script that applies a ratio and prints what the adapter reports, and gives the ratios measured on the wire at 100 kHz and 400 kHz with a logic analyzer.

1 Introduction

An I2C controller chooses a high time and a low time for every SCL period. The I2C specification sets minimums for both (tHIGH and tLOW), and most controllers pick a split on their own. That is fine until a target is sensitive to it, or until a test plan asks for a specific split: a 30:70 clock to check that a target samples correctly with a short high phase, or a 40:60 clock to match the timing of the system the target will ship in.

Until now the Binho adapters chose the split themselves. At 100 kHz the default is about 46 percent high; at 400 kHz, about 40 percent. There was no way to ask for another. Adapter firmware 4.6.0 and pyCosmicSDK 1.6.0 add one call that sets it.

1.1 Scope

This document covers:

- what `I2cController.set_duty_cycle` does and what it leaves alone
- the ratios firmware accepts, and what it answers for the ones it refuses
- why setting a ratio also makes the frequency exact
- a script that applies a ratio, generates traffic to measure, and prints what the adapter reports
- the split measured on the wire at 100 kHz and 400 kHz

It does not cover ratios outside 20 to 80 percent, duty-cycle control on the Supernova's I3C port, or SPI. SPI clock timing is a separate firmware change.

1.2 Applicable devices

Table 1. Applicable devices

Adapter	Identifier	Basis
Binho Supernova	hardware revision C, firmware 4.6.0 pre-release build <code>80cbb484</code>	measured, this document
Binho Pulsar	hardware revision C, firmware 4.6.0 pre-release build <code>ea77618b</code>	measured during firmware validation, within 1 percent of the Supernova; not re-measured for this document
Either adapter	firmware 4.5.1 or earlier	refuses the call with <code>FW_UNSUPPORTED_COMMAND</code>

Note: this revision was written against pre-release builds. The firmware reports its version as `4.5.1` with a build suffix because the version string had not yet been moved to 4.6.0. The call and its behavior are those of the 4.6.0 release.

1.3

Related documents**Table 2.** Related documents

Document	Title
UM10204	I2C-bus specification and user manual, Rev. 7, NXP Semiconductors
AN0014	Single-port I2C/SMBus/I3C operation for NVMe-MI on Binho Supernova

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System overview

`set_duty_cycle(high_pct)` sets the share of each SCL period the clock is held high. Three things follow from how the firmware implements it:

- **The frequency stays.** The bus keeps the frequency given to `bring_up` or `configure`. Only the split changes.
- **The frequency becomes exact.** Without a ratio, the adapter picks the nearest divider it can reach, and it may round above the request: 550 kHz runs at 600.6 kHz. With a ratio, the firmware moves the I2C peripheral onto a finer clock, so the frequency is never above the request and at most 0.29 percent below it. The same 550 kHz runs at 549.8 kHz.
- **The ratio persists.** A later `configure` keeps the ratio and applies it at the new frequency. `set_duty_cycle(0)` or a fresh `initialize` returns to the default timing.

The call returns what is in effect, as an `I2cDutyCycle` with two fields:

Table 3. Fields of `I2cDutyCycle`

Field	Meaning
<code>frequency_hz</code>	With a ratio set, the frequency on the wire. With <code>high_pct</code> 0, the requested frequency, which the default timing may round
<code>high_pct</code>	The ratio in effect, in percent; 0 is the default timing

The firmware does not clamp the ratio to the I2C specification. A 30 percent clock at 100 kHz has a `tHIGH` of 3.0 us, below Standard-mode's 4.0 us minimum. That is deliberate: out-of-spec timing is what margin testing needs. Check a ratio against the specification before using it in a product.

3 Hardware setup

3.1 Required equipment

Table 4. Required equipment

Item
Binho Supernova or Pulsar, firmware 4.6.0 or later
USB-C cable
An I2C target, or none if the goal is only to measure SCL
Oscilloscope or logic analyzer on SCL, to measure the result

3.2 Connections

Table 5. Connections to the I2C port

Adapter	Target or instrument	Required
SCL	target SCL, instrument probe	required
SDA	target SDA	required with a target
GND	target and instrument ground	required
3.3 V	target supply	optional

3.3 Bus considerations

The adapter measures t_{HIGH} from the moment it sees SCL high. A slow rising edge therefore lengthens the period: with a 2.2 kOhm pull-up, a 1 MHz clock runs at about 905 kHz. At 1 MHz use 470 Ohm or a stronger pull-up. At 100 kHz and 400 kHz, 1 kOhm to 2.2 kOhm is enough.

4 Software setup

Install pyCosmicSDK 1.6.0 or later:

```
pip install "pycosmicsdk>=1.6.0"
```

The script `i2c_duty_cycle.py` is in the assets archive. It needs nothing else.

5 The duty-cycle call

The call in isolation:

```

import pycosmicsdk as p

dev = p.Device.open(model=p.DeviceModel.SUPERNOVA)
bus = dev.i2c()
bus.set_voltage(voltage_mv=3300)
bus.bring_up(frequency_hz=400_000, pull_up=p.I2cPullUp.KOHM_1_0)
print(bus.set_duty_cycle(high_pct=40))
# I2cDutyCycle(frequency_hz=400000, high_pct=40)

```

Table 6. Values accepted by `set_duty_cycle`

high_pct	Result
0	default timing, the same as firmware 4.5.1
20, 30, 40, 50, 60, 70, 80	that share of each period high; frequency exact
anything else	<code>FirmwareError</code> , status <code>FW_INVALID_PARAMETER</code> ; the ratio in effect is kept

On firmware 4.5.1 and earlier, the call raises `FirmwareError` with status `FW_UNSUPPORTED_COMMAND`, and the bus keeps working.

6 Procedure

1. Connect the adapter and, if used, the instrument on SCL.
2. Ask the adapter what it reports for every ratio at the frequencies of interest. Each frequency is applied with `configure`, then every ratio in turn, then the default timing is restored:

```
$ python i2c_duty_cycle.py --model supernova --pull-up KOHM_1_0 table --khz 100 400 1000
SUPERNOVA FW-4.5.1-80cbb484 I2C 3300 mV pull-up KOHM_1_0
request 100 kHz -> 100000 Hz default
request 100 kHz -> 100000 Hz 20 % high
request 100 kHz -> 100000 Hz 30 % high
request 100 kHz -> 100000 Hz 40 % high
request 100 kHz -> 100000 Hz 50 % high
request 100 kHz -> 100000 Hz 60 % high
request 100 kHz -> 100000 Hz 70 % high
request 100 kHz -> 100000 Hz 80 % high
request 400 kHz -> 400000 Hz default
request 400 kHz -> 400000 Hz 20 % high
request 400 kHz -> 400000 Hz 30 % high
request 400 kHz -> 400000 Hz 40 % high
request 400 kHz -> 400000 Hz 50 % high
request 400 kHz -> 400000 Hz 60 % high
request 400 kHz -> 400000 Hz 70 % high
request 400 kHz -> 400000 Hz 80 % high
request 1000 kHz -> 1000000 Hz default
request 1000 kHz -> 999024 Hz 20 % high
request 1000 kHz -> 999024 Hz 30 % high
request 1000 kHz -> 999024 Hz 40 % high
request 1000 kHz -> 999024 Hz 50 % high
request 1000 kHz -> 999024 Hz 60 % high
request 1000 kHz -> 999024 Hz 70 % high
request 1000 kHz -> 999024 Hz 80 % high
```

At 1 MHz the exact frequency is 999024 Hz, 0.1 percent under the request. The `default` rows report the request, not the wire; section 8.2 shows the difference.

3. Apply one ratio. With `--traffic`, the script keeps SCL running for that many seconds so the instrument can capture it:

```
$ python i2c_duty_cycle.py --model supernova --pull-up KOHM_1_0 set --khz 100 --high 30
SUPERNOVA FW-4.5.1-80cbb484 I2C 3300 mV pull-up KOHM_1_0
request 100 kHz -> 100000 Hz 30 % high
```

4. A ratio outside the accepted set is refused, and the one in effect stays:

```
$ python i2c_duty_cycle.py --model supernova --pull-up KOHM_1_0 set --khz 100 --high 35
error: i2c_set_duty_cycle: FW_INVALID_PARAMETER: At least one of the parameters of the command is not valid.
SUPERNOVA FW-4.5.1-80cbb484 I2C 3300 mV pull-up KOHM_1_0
```

5. Compare a ratio with the default at a frequency the default rounds:

```
$ python i2c_duty_cycle.py --model supernova --pull-up KOHM_1_0 set --khz 550 --high 50
SUPERNOVA FW-4.5.1-80cbb484 I2C 3300 mV pull-up KOHM_1_0
request 550 kHz -> 549798 Hz 50 % high
$ python i2c_duty_cycle.py --model supernova --pull-up KOHM_1_0 set --khz 550 --high 0
SUPERNOVA FW-4.5.1-80cbb484 I2C 3300 mV pull-up KOHM_1_0
request 550 kHz -> 550000 Hz default
```

The second line reports 550000 Hz because it echoes the request. On the wire the default runs at 600.6 kHz (section 8.2).

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Reference

Table 7. i2c_duty_cycle.py options

Option	Default	Meaning
<code>--model</code>	any	<code>supernova</code> or <code>pulsar</code> , when both are connected
<code>--serial</code>	any	adapter serial number
<code>--voltage-mv</code>	3300	bus voltage
<code>--pull-up</code>	<code>KOHM_2_2</code>	on-board pull-up, any <code>I2cPullUp</code> name

Table 8. i2c_duty_cycle.py commands

Command	Options	Action
<code>set</code>	<code>--khz</code> , <code>--high</code> , <code>--traffic</code> , <code>--address</code>	apply one ratio; with <code>--traffic</code> , read from <code>--address</code> for that many seconds
<code>table</code>	<code>--khz</code> (one or more)	print the reported frequency and ratio for 0 and 20 to 80 percent, then restore the default

The traffic reads from address 0x50 by default. With no target there, each read is an address byte and a NACK, which still clocks nine SCL periods.

8 Measured results

8.1 Test configuration

Table 9. Test configuration

Item	Value
Adapter	Binho Supernova, hardware revision C, firmware FW-4.5.1-80cbb484 (4.6.0 pre-release)
Host software	pyCosmicSDK 1.6.0 pre-release, integration/1.6.0 at baacfdb8
Bus	I2C port, 3.3 V, 1 kOhm on-board pull-up, no target
Traffic	set --traffic 10 , about 4800 transfers per capture
Instrument	Acute TravelBus logic analyzer on SCL, 800 MS/s, threshold 1.6 V
Captures	3 per point, then 1 per point after a power cycle; about 15000 SCL periods each

8.2 Results

Median values. The ratio column is the median of tHIGH over the period across one capture; the p1 to p99 column is its spread. All four captures of each point agreed to 0.1 percent.

Table 10. SCL timing measured on the wire

Request	Ratio	Frequency	High	tHIGH	tLOW	High, p1 to p99
100 kHz	default	100.0 kHz	46.4 %	4645 ns	5360 ns	46.1 to 46.9 %
100 kHz	30	100.0 kHz	29.7 %	2975 ns	7030 ns	29.4 to 30.1 %
100 kHz	40	100.0 kHz	39.9 %	3990 ns	6015 ns	39.2 to 40.1 %
100 kHz	50	100.0 kHz	49.6 %	4965 ns	5035 ns	49.2 to 50.2 %
400 kHz	default	400.0 kHz	39.6 %	990 ns	1510 ns	38.6 to 41.1 %
400 kHz	30	400.0 kHz	28.0 %	700 ns	1800 ns	27.2 to 30.7 %
400 kHz	40	400.0 kHz	39.5 %	990 ns	1515 ns	38.8 to 40.7 %
400 kHz	50	400.0 kHz	48.8 %	1220 ns	1280 ns	48.2 to 50.7 %

At 400 kHz the measured high share sits 0.5 to 2 points under the ratio. The rising edge accounts for it: the analyzer reads SCL as high only once it crosses 1.6 V, and at 400 kHz that delay is a larger share of the period.

Two further points were measured during firmware validation on earlier builds of the same change, not with the configuration above:

- 1 MHz with a 470 Ohm pull-up, Supernova: 1000.0 kHz at 29.5, 37.0 and 50.5 percent for ratios 30, 40 and 50.
- 550 kHz: the default timing runs at 600.6 kHz at 50.2 percent high, on both adapters and on the 4.5.1 release. A ratio of 50 runs at 551.0 kHz.

The 20, 60, 70 and 80 percent ratios were checked through the adapter's report (section 6, step 2) and not on the wire.

9 Troubleshooting

Table 11. Troubleshooting

Symptom	Probable cause and action
<code>FW_UNSUPPORTED_COMMAND</code>	Firmware 4.5.1 or earlier. Update the adapter to 4.6.0 or later.
<code>FW_INVALID_PARAMETER</code>	The ratio is not 0 or 20 to 80 in steps of 10. The previous ratio is still in effect.
<code>FW_INTERFACE_NOT_INITIALIZED</code>	The I2C bus was not brought up as a controller. Call <code>bring_up</code> first.
Measured frequency below the request at 1 MHz	The rising edge is slow. Use a 470 Ohm or stronger pull-up (section 3.3).
Ratio back to default after a script restarts	<code>initialize</code> restores the default timing. Call <code>set_duty_cycle</code> after every <code>bring_up</code> that initializes the bus.
<code>CapabilityError</code> from <code>table</code>	<code>configure</code> needs a Supernova of hardware revision C. Use <code>set</code> once per frequency instead.

10 Acronyms

Table 12. Acronyms

Term	Meaning
I2C	Inter-Integrated Circuit
SCL	Serial clock line
SDA	Serial data line
tHIGH	High period of the SCL clock
tLOW	Low period of the SCL clock
NACK	Not acknowledge

11 Note about the source code in the document

Example code shown in this document and supplied in the accompanying archive is provided for evaluation and reference. It is released under the MIT license, the text of which is included in the archive.

The archive `AN0015-assets.zip` contains `i2c_duty_cycle.py` and `LICENSE`.

12 Revision history

Table 13. Revision history

Revision	Date	Description
0.1	25 September 2026	Draft for internal review, measured on firmware 4.6.0 and pyCosmicSDK 1.6.0 pre-release builds.

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